Memory Forensics Investigation Report

This project focuses on analyzing a memory image using Volatility to identify potential malicious activity and understand what was happening on the system at the time of capture.

Objectives
Identify running processes, detect suspicious activity, analyze network connections, and determine indicators of compromise.

Tools Used
Volatility, Volatility 3, Python

Methodology

```
volatility -f memory.raw windows.pslist
volatility -f memory.raw windows.netscan
volatility -f memory.raw windows.cmdline
volatility -f memory.raw windows.malfind
```

Analysis
Processes were reviewed to identify anomalies such as unusual parent-child relationships and unexpected command execution behavior. Network connections were analyzed to detect suspicious external communication.

Findings
Suspicious processes and possible injected code regions were identified, suggesting potential compromise.

Conclusion
The analysis indicates signs of malicious activity within the memory image. Further investigation and system hardening would be recommended.